Miguel Monteiro

Mechanical Engineer

Portfolio: GitHub Pages
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Education

MSc in Mechanical Engineering, Technical University of Munich — *Present*
BSc in Mechanical Engineering, Faculty of Engineering of the University of Porto — *Sep. 2022 - Jul. 2026*

Work Experience

Formula Student FEUP — Porto, Portugal

Head of Chassis — *Jan. 2025 – Jul. 2026*

- Managed **3 subsystems** and **8 people**;
- Planned the team's **first monocoque** as a proof-of-concept;
- Designed the third spaceframe achieving a mass of 29 kg, a **24% decrease**;
- Presented all static disciplines in 2025 and 2026 competitions, achieving **12 podiums** out of 15, including the **first Portuguese podium at Formula Student Germany** in the **EV** category;
- **Co-supervised** 2 Master and 1 Bachelor thesis about **composite design** and **FEA simulations**.

Primary Structure Manufacturing Responsible — *Sep. 2024 – Jan. 2025*

- Coordinated the chassis' production and obtained **6 new partners** to ensure project feasibility;
- Designed the needed infrastructure to guarantee welding tolerance.

Primary Structure Engineer — *Nov. 2023 – Sep. 2024*

- Designed the spaceframe and achieved a mass of 38 kg, a **25% reduction**;
- Created a **topology optimized** steering support using SLM, achieving a **55% reduction** from its machined solution;
- Modeled and sized all primary mounts using both **analytical and computational methods**;
- **Co-coordinated** a Bachelor thesis about the **mechanical characterization of sandwich panels**.

Adamastor Supercars — Porto, Portugal

Research Internship — *Jan. 2025 – Jun. 2025*

- Bachelor thesis about the influence of inserts on the stiffness of sandwich panels, achieved a **grade of 18/20**;
- **Developed** a **numerical sizing tool** for through-the-core inserts;
- Performed **linear simulations** to predict the **elastic response** of **sandwich panels**;
- Planned, manufactured and tested specimens under **four-point bending**.

Additional Experience

NEEM FEUP — Porto, Portugal

Marketing Member — *Feb. 2023 – Feb. 2024*

- Responsible for clothing design and manufacture;
- Photography and creation of media content for the National Meeting of Mechanical Engineers.

Skills

Languages	**Portuguese** (native), **English** (C2, proficient), **German** (B1, intermediate)
FEA Software	Abaqus, Ntopology
CAD Software	Siemens NX, Dassault Solidworks, Dassault Catia, Autodesk Inventor, Autodesk Fusion
Programming	Matlab, Python, Javascript
Management	Notion, Gantt